Experiment No-12 : Tic Tac Toe Game

Aim

To write a Python program to implement the Tic Tac Toe game.

Programming Code:

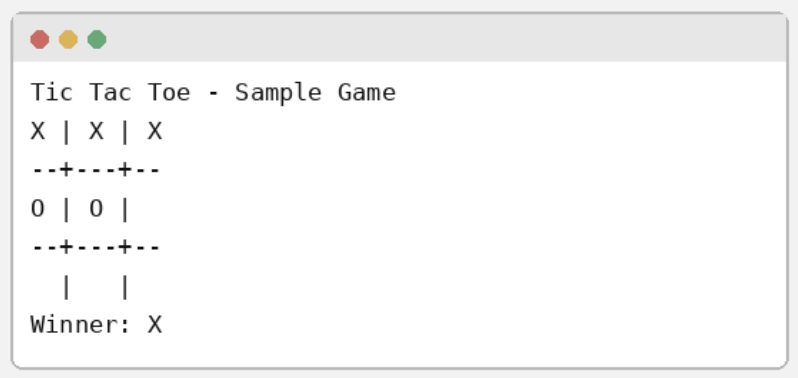
```
def print_board(board):
    for i in range(0, 9, 3):
        print(" | ".join(board[i:i + 3]))
        if i < 6:
            print("--+---+--")

def winner(board, mark):
    wins = [(0,1,2), (3,4,5), (6,7,8),
            (0,3,6), (1,4,7), (2,5,8),
            (0,4,8), (2,4,6)]
    return any(all(board[i] == mark for i in line)
               for line in wins)

board = [' '] * 9
moves = [0, 4, 1, 3, 2]  # sample moves: X wins
for turn, move in enumerate(moves):
    mark = 'X' if turn % 2 == 0 else 'O'
    board[move] = mark
    if winner(board, mark):
        break

print("Tic Tac Toe - Sample Game")
print_board(board)
if winner(board, mark):
    print("Winner:", mark)
elif ' ' not in board:
    print("Result: Draw")
else:
    print("Game in progress")
```

OUTPUT:



Result

Thus, the Tic Tac Toe game was implemented successfully in Python. The program displays the board and checks the winning combinations to identify the winner.